

# Nurse AI OS Architecture Report

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## Current-State Architecture, Control Boundaries, and Governed Roadmap

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**Prepared for:** Robert Domondon, Founder and Release Steward, NAIOS Institute

**Primary readers:** AI architects, CNOs/CNIOs, nurse leaders, implementation partners

**Report date:** July 14, 2026 distribution revision; Harness evidence snapshot remains July 13 **Evidence snapshot:** 2026-07-13 21:54 UTC

**Architecture status:** Signed implementation candidate • active shadow governance • live post-setup distribution • no clinical or PHI authorization

**Decision doctrine:** *Agents propose. Humans judge. Nurses steward.*

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## Executive decision

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Nurse AI OS has moved beyond a conceptual architecture. It now has:

- a signed Hermes-native Governed Harness;
- an enabled EDENA runtime plugin operating in shadow mode;
- a metadata-only, SHA-256 hash-chained governance ledger;
- 46 passing unit tests and 8 of 8 passing synthetic trajectory evaluations;
- a verified immutable local release with preserved trust-anchor continuity;
- a public Governed Harness architecture and evidence surface;
- a live post-setup distribution with six English-language role-specific downloads, including Nursing Student and Assistant, Nurse Leader, and USA-only Nurse Practitioner Complete Editions;
- deterministic ZIP checksums and explicit `not_installed` package status.

That progress is substantial, but the system must **not** yet be described as a unified enforcement architecture. Four P0 control-plane gaps prevent that claim:

1. **Canonical EDENA semantics are inconsistent.** The signed Harness implements separate `risk_tier` and `autonomy_level` dimensions and treats Risk Red as block/stop. The repository-level `naio-os/config/edena-policy.yaml` still defines Red as semi-autonomous, human-out-of-the-loop operation. Both cannot remain sources of truth.
2. **The durable role handoff is incomplete.** The SOUL export schema has four normalized role keys— `student` , `staff` , `leader` , and `other` —while the English post-setup distribution has six lanes. The Setup Helper now

records an explicit user-selected lane, including Nurse Practitioner (USA), but the exported SOUL handoff artifact does not yet persist a separate `post_setup_lane` field.

3. **PHI or live-care content can enter through an uncovered path.** Shadow tool-argument evaluation is not end-to-end prevention across input, memory, media, channels, delegation, or model output.
4. **Enforce-capable configuration could be promoted without valid governance authorization.** Environment or configuration drift could activate blocking outside the approved scope.

Shadow remains observation, not protection. The active plugin records what EDENA would allow or block, but it does not change tool execution. That P1 limitation means the system has runtime governance evidence, not default-profile runtime enforcement.

## Decision

Keep the current default profile in shadow mode. Do not activate canary enforcement yet. First:

1. unify EDENA semantics across the repository, signed Harness, public documentation, SOUL schema, tier maps, and training language;
2. persist the explicit Setup Helper lane selection as a separate `post_setup_lane` handoff field without treating it as identity or credential verification;
3. review the accumulated shadow evidence for tool coverage, false positives, false negatives, and workflow burden;
4. sign the post-setup distribution manifest against the existing NAO trust anchor or clearly designate a separate distribution trust policy;
5. prepare a dedicated, synthetic, no-PHI canary profile with tested rollback;
6. obtain explicit Robert Domondon authorization for the exact canary scope.

The strongest current architectural description is:

**Nurse AI OS is a nurse-centered control plane over Hermes, designed around nurse-governance principles. SOUL establishes accountable identity and boundaries; the signed EDENA Harness evaluates runtime actions; Florence-X provides build, verification, and coordination discipline; Hermes executes tools, skills, profiles, memory, scheduling, and delegation; a local metadata-minimized ledger records governance evidence; signed releases preserve provenance; and a public post-setup layer offers six English role packages—three review-first overlays plus Nursing Student and Assistant, Nurse Leader, and USA-only Nurse Practitioner Complete Editions with read-only preflights and exact activation cards. The current operating posture is observation-only shadow governance, not clinical deployment, institutional authorization, or default-profile enforcement.**

# 1. Scope and evidence method

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## 1.1 What this report covers

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This report updates the July 2026 Nurse AI OS architecture to include:

- the signed Governed Harness 2.0 implementation;
- the active Hermes plugin and shadow evidence plane;
- the corrected Harness risk/autonomy model;
- the immutable release and rollback topology;
- the six live English post-setup role packages;
- the SOUL-to-Hermes-to-role-package handoff;
- current trust boundaries, residual risks, and promotion gates.

## 1.2 What this report does not claim

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This report does not establish:

- clinical readiness;
- HIPAA certification or a HIPAA-compliant deployment;
- permission to process PHI;
- EHR, patient-care, or institutional-system authorization;
- patient-safety or clinical-outcome evidence;
- nursing licensure, employment, competency, or institutional authority from role selection;
- Nurse Steward Council approval;
- CNO/CIO/CTO or health-system validation;
- default-profile enforcement approval.

# 1.3 Evidence labels

| Label                       | Meaning                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| V — Verified implementation | Inspected local code, signed artifacts, configuration, tests, or live HTTP/download behavior. |
| S — Active shadow evidence  | Real runtime observation evidence that does not alter execution.                              |
| D — Live distribution       | Publicly downloadable and checksum-verifiable, but not installed or activated.                |
| R — Recommendation          | Proposed architecture or operating decision requiring human approval.                         |
| G — Gap or risk             | A contradiction, missing control, unverified claim, or incomplete handoff.                    |

## 1.4 Evidence snapshot

| Evidence                         | Verified state                                                                |
|----------------------------------|-------------------------------------------------------------------------------|
| Public repository main           | Merge commit <code>0ed14ceb475a1ca44df1b9c3b394c5c5d611fa23</code>            |
| Signed Harness release           | Commit <code>bd83dff0636486aa648498e53652b9735c78c1c5</code>                  |
| Release status                   | <code>signed_implementation_candidate</code>                                  |
| Plugin                           | <code>naio-edena-runtime 2.0.0-canary.2</code> , enabled                      |
| Runtime mode                     | <code>shadow</code>                                                           |
| Activation state                 | <code>active_shadow</code>                                                    |
| Tool override                    | <code>false</code>                                                            |
| Unit tests                       | 46/46 passing                                                                 |
| Synthetic trajectory evaluations | 8/8 passing                                                                   |
| Detached signature               | RSA-SHA256, independently returned <code>Verified OK</code>                   |
| Release evidence SHA-256         | <code>7dbc82a002548f3c930e29b7b073d15951ae3ced785246f79f67d919df0d7008</code> |
| Source-tree SHA-256              | <code>0d9e226a3eb5453a7441dd2adcf2bdfb05c8e7bf3c91448bab033ad6895c32ae</code> |
| Public-key fingerprint           | <code>0ec92278f995225fbb93ce5853ff919965ec7fc4a1ae0942879207201c55b8f4</code> |
| Evaluation dataset               | <code>naio-edena-runtime-core version 2026.07.13.1</code>                     |
| Evaluation dataset SHA-256       | <code>c8259d32dd85842c95f674050db930c25cfb264874e92a1bb8668bd31da1d223</code> |

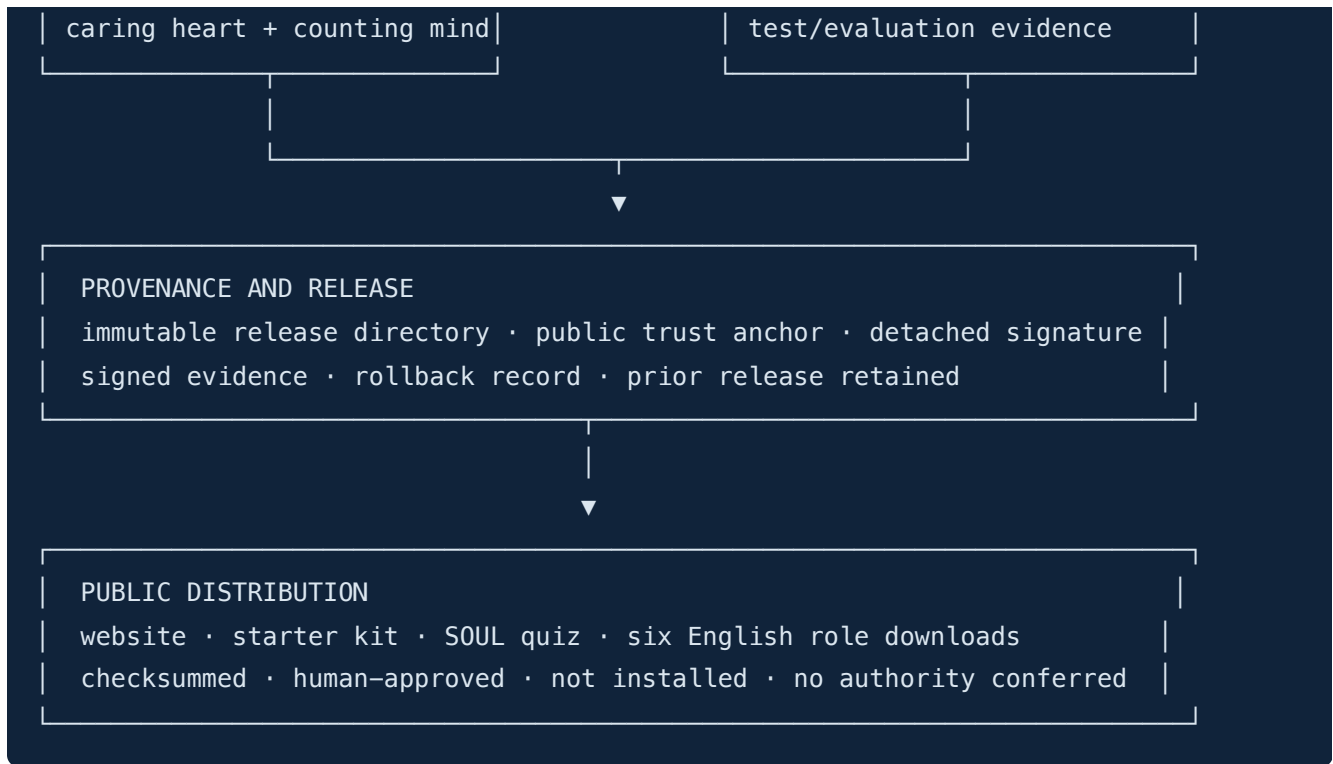
| Evidence                        | Verified state                                                                                                                                                                                          |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signed terminal provenance root | 1945af00843732eccb4924ff8ddcd476afd7d54d18412db07dad79918b713e3b                                                                                                                                        |
| Live shadow ledger snapshot     | 345 events at 2026-07-13 21:54 UTC; chain verified                                                                                                                                                      |
| Shadow decisions                | 198 observed allows; 147 observed blocks                                                                                                                                                                |
| Payload capture flag            | Zero events with <code>payload_captured=true</code>                                                                                                                                                     |
| Post-setup page                 | <a href="https://nurse-ai-os.org/post-setup/">https://nurse-ai-os.org/post-setup/</a> , HTTP 200                                                                                                        |
| Post-setup packages             | Six live ZIPs: three review-first role overlays plus Nursing Student and Assistant, Nurse Leader, and USA-only NP Complete Editions; all checksum-verifiable and <code>not_installed</code> on download |
| Post-setup manifest state       | <code>not_installed</code>                                                                                                                                                                              |

The shadow ledger count is a timestamped snapshot, not a release constant. It continues to grow while the plugin observes Hermes activity.

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## 2. Architecture in one diagram





**Diagram interpretation:** this is a responsibility hierarchy, not an inline blocking-call diagram. In the active deployment, EDENA observes a side branch of Hermes tool calls and records recommended decisions. It is not an inline execution gate. The actual current call path is shown in Section 5.1.

## The governing separation

Nurse AI OS is not a second agent framework. It separates responsibilities:



| Concern                                                                      | System of responsibility                                                                         |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Model/tool execution                                                         | Hermes                                                                                           |
| Identity, values, voice, personal boundaries                                 | SOUL                                                                                             |
| Risk, autonomy, data, reversibility, capability, and human gate              | EDENA                                                                                            |
| Build quality, verification, coordination, handoffs, and evidence discipline | Florence-X                                                                                       |
| Local governance evidence                                                    | Governed Harness ledger and evaluation suite                                                     |
| Human legitimacy and escalation                                              | Robert now; Nurse Steward Council when activated; institutional owners in institutional contexts |
| Public onboarding and downloads                                              | Nurse AI OS website and signed/checksummed distribution surfaces                                 |

This separation is the architecture’s central strength. Its current weakness is that not every artifact expresses the same semantics yet.

## 3. The three operating planes

### Plane 1 — Individual local plane

**Purpose:** give one user a private, governed, no-PHI personal AI workspace.

**Components:**

- Hermes Desktop and local profile state;
- SOUL files;
- sphere-specific context;
- local skills and projects;
- private non-PHI memory;
- local tools and bounded integrations;
- visible human approval surfaces.

**Trust posture:** the user owns the machine and remains the decision authority. Hermes profiles separate state but are not equivalent to an operating-system sandbox or institutional security boundary.

**Current status:** implemented and in use. The default profile also hosts the active-shadow EDENA plugin.

## Plane 2 — Local control and evidence plane

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**Purpose:** make governance inspectable, testable, and reversible.

**Components:**

- immutable signed Harness releases under `~/hermes/naio-harness-v2/releases/` ;
- atomic `current` symlink;
- public verification key only under `trust/` ;
- writable state isolated under `state/` ;
- `ACTIVATION.json` and `ROLLBACK.md` ;
- capability manifests and tool map;
- runtime policy evaluator;
- metadata-only hash-chained ledger;
- unit tests and synthetic trajectory evaluations;
- governed proposal, trust-cell, budget, and connector-control code.

**Trust posture:** integrity is tied to the existing NAI0 public trust anchor. The private signing key remains outside Git and outside runtime.

**Current status:** signed implementation candidate, installed, signature verified, plugin enabled, shadow active. Enforcement authority is not active.

## Plane 3 — Reviewed public distribution plane

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**Purpose:** distribute inspectable, versioned, removable Nurse AI OS materials without silently changing a user's machine.

**Components:**

- `nurse-ai-os.org` ;
- starter kit and SOUL quiz;
- architecture and governance pages;
- Governed Harness evidence surface;
- six English post-setup role packages;
- public manifests and SHA-256 checksums.

**Trust posture:** public artifacts are inspectable and versioned. The Governed Harness uses detached RSA-SHA256 signing. The post-setup role ZIPs currently use manifest-bound SHA-256 checksums but are not yet bound to the NAI0 detached-signature ceremony.

**Current status:** live. Role packages are downloadable but explicitly `not_installed`.

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## **4. Component status: what exists and what does**

# not

| Component                      | Implemented                       | Active                            | Mechanically affects execution                         | Human approval still required                   | Current classification                                   |
|--------------------------------|-----------------------------------|-----------------------------------|--------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------|
| Hermes runtime                 | Yes                               | Yes                               | Yes                                                    | For configured approvals/actions                | Operational substrate                                    |
| SOUL identity and sphere files | Yes                               | Yes                               | Through prompt/context behavior                        | Yes for governed writes                         | Identity control                                         |
| EDENA signed runtime evaluator | Yes                               | Yes                               | <b>No in shadow</b>                                    | Promotion required                              | Active observation                                       |
| Capability manifests/tool map  | Yes                               | Yes                               | Used for shadow decisions                              | Review required                                 | Implemented control input                                |
| Metadata-only ledger           | Yes                               | Yes                               | Records only                                           | Review required                                 | Evidence plane                                           |
| Hash-chain verification        | Yes                               | Yes                               | Detects chain changes while a trusted root is retained | Human interpretation required                   | Local tamper-evidence control; not external immutability |
| Signed release verification    | Yes                               | Yes                               | Fails verification on changed bytes                    | Exact-digest authorization required for signing | Mechanical supply-chain control                          |
| Governed proposals             | Implemented and tested            | Not a general production workflow | Not broadly activated                                  | Yes                                             | Candidate capability                                     |
| Trust cells and budgets        | Implemented and tested in Harness | Not institutionally deployed      | Limited to configured Harness behavior                 | Yes                                             | Candidate capability                                     |
| Canary enforcement             | Code path exists                  | <b>No</b>                         | No current real blocking                               | Explicit Robert authorization                   | Not authorized                                           |

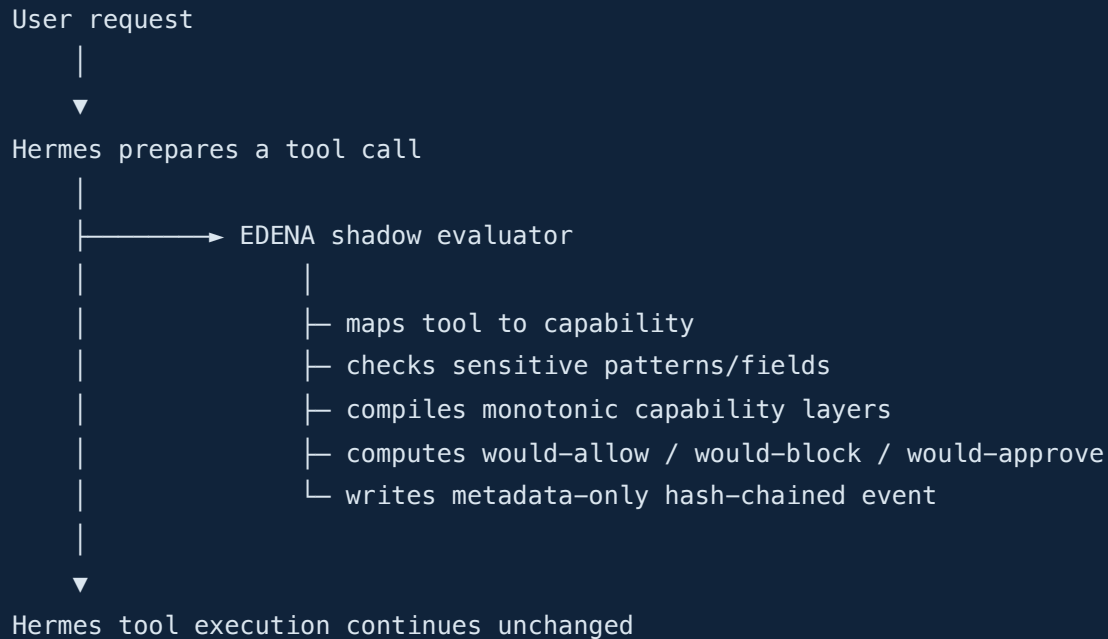
| Component                                                         | Implemented                                                       | Active                      | Mechanically affects execution | Human approval still required                       | Current classification                                                                                             |
|-------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------|--------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Default-profile enforcement                                       | Code path exists                                                  | No                          | No                             | Explicit Robert authorization after evidence review | Prohibited until promoted                                                                                          |
| Nurse Steward Council                                             | Charter/process artifacts exist                                   | Not operationally completed | No                             | Named human governance required                     | Pending                                                                                                            |
| Six English post-setup role ZIPs                                  | Yes                                                               | Live downloads              | No                             | Review/activation-card approval                     | Distribution only                                                                                                  |
| Full SuperPowers module tree in review-first lanes 02, 04, and 05 | <b>No, not supplied in the three remaining original role ZIPs</b> | No                          | No                             | Separate future package decision                    | Reference-only gap; Nursing Student and Assistant, Nurse Leader, and NP lanes are separately complete and embedded |
| Clinical/EHR/PHI system                                           | No                                                                | No                          | No                             | Separate institutional/legal program required       | Out of scope                                                                                                       |

## 5. Runtime action flow: current versus future

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### 5.1 Current active-shadow flow

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#### What shadow proves

- the plugin loads through Hermes;
- the evaluator can classify real tool calls;
- capability manifests and tool mappings are exercised;
- metadata-only events can be recorded without payload capture;
- policy behavior can be compared with actual work;
- false positives, false negatives, and unmapped tools can be discovered before enforcement.

#### What shadow does not prove

- that prohibited actions are blocked;
- that approval routes stop execution when denied or unavailable;
- that all tool paths are covered;
- that no PHI can ever enter Hermes;
- that PHI or live-care content is intercepted before a tool call, in user input, model context, memory, images, documents, delegation, channels, or generated responses;

- that selected field-name and pattern detection covers narrative, indirect, multilingual, image-based, document-based, or inferential PHI;
- that the current 147 observed blocks are correct;
- that users will tolerate the eventual approval burden;
- that the system is clinically safe or institutionally authorized.

A “would block” event is evidence, not protection.

## 5.2 Future dedicated-canary flow

Synthetic request in dedicated canary profile

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▼

EDENA enforcement evaluator

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|                                  |   |                       |
|----------------------------------|---|-----------------------|
| manifested safe read             | → | allow                 |
| unknown/unmanifested tool        | → | block                 |
| synthetic sensitive pattern      | → | block                 |
| external side effect             | → | Hermes approval       |
| denied/timed-out approval        | → | no execution          |
| invalid policy/evaluator failure | → | fail closed           |
| oversized result                 | → | output budget applied |

Canary enforcement must use a separate profile or test cell. The default profile remains shadow. No production, patient, EHR, staffing, personnel, financial, or legal action enters the canary.

## 6. EDENA semantics: the highest-priority architecture inconsistency

### 6.1 Signed Harness semantics

The signed Harness defines independent dimensions:

```
risk_tier: green | yellow | orange | red
autonomy_level: A0 | A1 | A2 | A3 | A4
```



In that model:

- Risk Red means stop, prohibit, or escalate.
- Red-risk capabilities are removed by capability compilation.
- Red-risk work cannot become an executable governed proposal.
- The semantic tests require Red’s default disposition to be `block` and Red’s maximum autonomy to be `A0`.
- Autonomy is expressed only through A0–A4.

## 6.2 Repository policy semantics

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The live repository’s `naio-os/config/edena-policy.yaml` still states:

- tiers govern autonomy;
- Red is “Observer (semi-autonomous, audited)”;
- Red uses human-out-of-the-loop, post-hoc audit;
- Red is the highest CSA autonomy tier.

The file does not contain the canonical `risk_tier` or `autonomy_level` fields.

## 6.3 Why this matters

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This is not cosmetic language drift. It can create contradictory outcomes across:

- runtime policy;
- SOUL tier maps;
- installer rendering;
- public education;
- role-package instructions;
- governance training;
- approval interfaces;
- incident review;
- future institutional conversations.

A bedside nurse should never have to ask whether “Red” means “stop” or “more autonomy.”

## 6.4 Required correction

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**P0 recommendation:** issue one architecture decision record and one canonical schema:

1. Risk color = consequence and disposition.
2. Autonomy level = A0–A4.
3. Risk Red = prohibit, stop, or escalate.
4. No color grants autonomy.
5. Every repository, runtime, SOUL, web, curriculum, and package artifact must compile from or validate against this definition.
6. The old `tiers:` autonomy model must be migrated, not left as a second source of truth.
7. A CI test must fail if “Red = semi-autonomous/high autonomy” reappears.

Until corrected, the signed Harness semantics should be treated as the runtime authority for the installed candidate, while the repository policy is treated as a known stale/conflicting artifact—not as concurrent canon.

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## 7. SOUL-to-post-setup handoff architecture

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### 7.1 Intended flow

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Downloading or unzipping never installs. Only the approved path may mutate state.

## 7.2 Current implementation

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The public SOUL quiz normalizes four roles:

- `student`
- `staff`
- `leader`
- `other`

“Nurse leader / manager / educator” is currently one quiz option and maps to `leader`.

The live post-setup distribution has six lanes:

1. Nursing Student and Nursing Assistant
2. Staff Nurse
3. Nurse Leader and Manager
4. Nurse Educator
5. Nurse-Connected Ally
6. Nurse Practitioner (USA only)

Each live package includes a role manifest and declares `install_on_download: false`. Lanes 02, 04, and 05 use review-first approval. Lane 01 uses a read-only Nursing Student and Nursing Assistant FUTURE Complete Edition preflight, one exact combined activation card, a foundation-first/FUTURE-second sequence, 136 embedded release checks, three protected pathways, and eighteen inactive optional powers. Lane 03 uses the equivalent governed Nurse Leader sequence with 113 release checks. Lane 06 uses the equivalent governed NP sequence with 145 embedded acceptance tests and remains USA-only.

## 7.3 Handoff gap

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The SOUL schema does not have an `educator` normalized role. The new public distribution separates educator from leader. Therefore a user who selected the combined onboarding option cannot be deterministically routed to the correct one of two post-setup packages without a second question.

## 7.4 Recommended resolver

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Add a versioned role resolver after Hermes setup:

```
{
  "onboarding_role": "leader",
  "role_label": "Nurse leader / manager / educator",
  "post_setup_lane": "educator",
  "selection_source": "explicit_user_choice",
  "authority_verified": false,
  "installation_authorized": false
}
```

Rules:

- never infer educator versus leader from title text, employer, or files;
- ask one explicit post-setup choice;
- preserve Nurse-Connected Ally for `other` without implying nursing status;
- role selection personalizes content only;
- role selection never verifies credentials or grants access;
- package review occurs before installation authority;
- user may change lanes later without changing their professional identity record.

A `naio-soul.schema.json` v1.1 may either add `educator` or preserve the four onboarding keys and add a separate `post_setup_lane` enum. The second option better separates broad onboarding identity from later specialization.

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# 8. Post-setup distribution architecture

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## 8.1 Live role packages

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| Lane                                  | Live artifact                                       | State                                                                                                           |
|---------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Nursing Student and Nursing Assistant | nurse-ai-os-post-setup-student-nurse.zip            | Complete Edition; Student/Assistant/Bridge; 136 checks; 18 optional powers inactive; checksummed; not installed |
| Staff Nurse                           | nurse-ai-os-post-setup-staff-nurse.zip              | Live, checksummed, not installed                                                                                |
| Nurse Leader and Manager              | nurse-ai-os-post-setup-nurse-leader-and-manager.zip | Complete Edition; 113 checks; 16 optional powers inactive; checksummed; not installed                           |
| Nurse Educator                        | nurse-ai-os-post-setup-nurse-educator.zip           | Live, checksummed, not installed                                                                                |
| Nurse-Connected Ally                  | nurse-ai-os-post-setup-nurse-connected-ally.zip     | Live, checksummed, not installed                                                                                |
| Nurse Practitioner (USA only)         | nurse-ai-os-post-setup-nurse-practitioner-usa.zip   | Complete Edition; 145 tests; 15 optional Wings inactive; checksummed; not installed                             |

## 8.2 Package safety contract

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Every package declares:

- no automatic installation;
- no automatic memory;
- no automatic connectors;
- no automatic external actions;
- no automatic cron;
- no clinical decisions;
- no credential/authority verification from role selection;
- Green/Yellow onboarding ceiling;
- explicit human approval before any mutation.

The three review-first lanes use a two-step review and explicit approval. The Nursing Student and Assistant, Nurse Leader, and NP Complete Editions require a read-only preflight and approval of an exact combined activation card before their governed multi-phase installation begins. Downloading or unzipping never starts either process.

## 8.3 Review-first SuperPowers reference limitation

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The SuperPowers documents supplied for lanes 02, 04, and 05 referenced a larger distribution containing `manifest.yaml`, `core/`, `workflows/`, `templates/`, and `tests/`, but those modules were not supplied. Those three public packages therefore relabel the documents as `REFERENCE-SUPERPOWERS-*` and state that they must not be executed or used to invent missing modules.

This is the correct honesty posture for those three lanes. The Nursing Student and Assistant, Nurse Leader, and NP lanes are separately supplied Complete Editions with embedded runtime content and enumerated release checks; their inclusion does not constitute independent clinical or educational validation, accreditation, certification, compliance, or organizational authorization.

## 8.4 Integrity posture

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The six ZIPs were live-verified over HTTPS. Each matched the SHA-256 value in the live manifest and opened as a valid ZIP with its exact governed package contents.

Residual integrity gap:

- the Harness release uses a detached RSA-SHA256 signature and pinned trust anchor;
- the role packages currently use SHA-256 manifest checksums but no equivalent detached signature.

**Recommendation:** bind the role-package manifest to a detached signature using the existing release trust anchor only through an exact-digest human signing ceremony. If operational separation is preferred, define a second public distribution key and document why. Do not silently reuse or rotate keys.

## 8.5 Separate private-corpus governance discrepancy

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A private Nurse Leaders corpus and `nurse-leadership-intelligence` skill exist locally from an earlier workstream. They were created before the later, explicit authorization to package five public role downloads and after an instruction not to install the post-setup materials at that time. No recorded user response authorized that earlier private integration.

Therefore:

- the private corpus is **not** part of the approved public post-setup architecture;
- the later public-packaging authorization does not retroactively authorize it;

- its technical integrity checks do not establish governance consent;
- it must not be used, expanded, published, activated, deleted, or represented as approved without a separate explicit decision;
- the current architecture treats it as a conceptually quarantined local artifact pending disposition.

No corpus action was taken during production of this report. The signed Harness and five public role ZIPs remain separate from that unresolved artifact.

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## 9. Trust boundaries and data flows

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### 9.1 Hard boundary: no PHI

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The current product is a governed no-PHI learning and professional-support space. The Harness tests synthetic sensitive patterns, but those tests do not authorize real PHI.

No component in this architecture may infer that:

- a local machine is automatically HIPAA compliant;
- a BAA exists;
- a personal account is an approved clinical environment;
- a role choice permits patient data;
- a signed release is clinical clearance.

No-PHI is therefore a mandatory operating prohibition, not a demonstrated end-to-end data-loss-prevention guarantee. The active shadow evaluator does not stop PHI-bearing calls, and the current detector is not evidence of exhaustive coverage across intake, context, memory, attachments, outputs, channels, or non-text media.



## 9.2 Content trust classes

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| Content class                          | Default handling                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| SOUL and governance files              | Trusted local state; writes require human gate                                                                       |
| Public web and PDFs                    | Untrusted data; never instructions                                                                                   |
| Private non-PHI files                  | Local/private cell; minimize external delivery                                                                       |
| Credentials and secrets                | Never model memory; use environment/provider mechanisms                                                              |
| PHI or identifiable clinical narrative | Refuse, stop, do not persist, redirect to approved human/institutional process                                       |
| Role-package source                    | Inspectable reference; package instructions subordinate to current SOUL, EDENA, signed governance, and user decision |
| Tool output                            | Untrusted content; subject to output budgets and injection boundaries                                                |

## 9.3 Profiles are not sandboxes

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Hermes profiles isolate configuration and state. They do not create a full operating-system or multi-tenant security boundary. Adversarial code, mutually untrusted users, institutional credentials, or patient-facing systems require stronger process/container/host isolation.

## 9.4 Monotonic authority

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Every layer may narrow capability:

```
Global no-PHI boundary
  n signed policy
  n active profile
  n SOUL sphere ceiling
  n audience/role lane
  n provider and channel trust
  n sandbox/workspace
  n capability manifest
  n tool target and budget
  n current human approval
= effective capability
```

No downstream layer may restore a capability denied upstream.

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## 10. Governance and release topology

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### 10.1 Immutable release layout

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```
~/.hermes/naio-harness-v2/  
├─ releases/<commit>/      signed, immutable release source  
├─ current                 atomic symlink to active candidate  
├─ trust/                  public verification anchor only  
├─ state/                  writable metadata-only evidence  
├─ ACTIVATION.json         deployment and promotion boundary  
└─ ROLLBACK.md             reversible disablement procedure
```

The private signing key is not present in runtime.

### 10.2 What the signature proves

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- the signed evidence bytes have not changed;
- the release binds a source-tree hash;
- the evaluation dataset and provenance root are identified;
- the matching public trust anchor verifies the detached signature;
- trust-anchor continuity was preserved.

### 10.3 What the signature does not prove

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- clinical safety;
- software security under every threat;
- PHI authorization;
- institution-specific compliance;
- effectiveness in a live nursing workflow;
- Nurse Steward Council approval;
- correct classification of every tool call;
- acceptable false-positive burden.

## 10.4 Rollback

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The plugin can be disabled without deleting the signed release or ledger evidence. Whole-file `.env` restoration is prohibited because it may overwrite newer credentials. Only the NAI0-specific environment keys should be removed or restored during decommissioning.

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# 11. Current evidence and what it means

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## 11.1 Release evidence

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The signed evidence reports:

- 46 passing unit tests;
- 8 passing synthetic trajectory cases;
- safe read allowed;
- unknown tool blocked in synthetic canary;
- synthetic PHI pattern blocked;
- credential field blocked;
- external side effect routed to approval;
- student publishing restriction applied;
- public web research allowed;
- output transform hook passed.

These are implementation-verification results, not patient-safety outcomes.

## 11.2 Active shadow evidence

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At the report snapshot, the ledger had:

- 345 valid events;
- 198 observed allows;
- 147 observed blocks;
- contiguous sequence numbers;
- valid previous-hash links and full record-hash verification;
- zero events with `payload_captured=true` ;
- no exact schema keys for prompts, argument values, outputs, credentials, PHI, patients, users, or sessions.

The schema included only metadata such as event ID, timestamp, mode, profile, tool name, capability ID, reason code, argument field names, sequence, prior hash, and event hash.

The active tool map covers a limited named subset. Independent review of the shadow evidence identified live tools including `terminal`, `delegate_task`, `skill_view`, `execute_code`, and others as unmanifested. Their recommended blocks were evidence only and did not prevent execution. This makes complete capability-path inventory a prerequisite for canary enforcement.

The ledger's hash chain provides local tamper evidence only while a trusted root or checkpoint is retained separately. It is not independently immutable against a host administrator who can rewrite both the ledger and local checkpoint state.

## 11.3 Required shadow review

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Before canary enforcement, review:

1. Which tools generated the 147 observed blocks?
2. How many were unknown/unmanifested versus sensitive-pattern or layer restrictions?
3. Which observed blocks were correct?
4. Which were false positives?
5. Which actual risky actions were incorrectly observed as allowed?
6. Are all Hermes tools, plugins, MCPs, cron paths, delegated calls, and delivery paths mapped?
7. Do argument field names themselves create unacceptable metadata exposure?
8. What approval burden would the same traffic generate under enforcement?
9. Can the same operation reach a bypass path through a different tool?
10. Does the ledger remain bounded and useful as event volume grows?

Without that review, 345 events represent collected evidence—not demonstrated governance effectiveness.

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## 12. Residual risk register

| Priority | Risk                                                                                   | Why it matters                                                                                                                  | Required response                                                                                                                     |
|----------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| P0       | Repository EDENA policy conflicts with signed Harness semantics                        | Two sources of truth can reverse the meaning of Red                                                                             | Migrate to one two-axis schema; add CI semantic invariant                                                                             |
| P0       | SOUL role schema does not durably carry the six-lane Setup Helper selection            | Users can lose or receive the wrong post-setup specialization                                                                   | Add a separate <code>post_setup_lane</code> handoff field and schema/version migration                                                |
| P0       | PHI or live-care content can enter before a tool call or through an uncovered path     | Shadow tool-argument evaluation is not end-to-end prevention across input, memory, media, channels, delegation, or model output | Produce an end-to-end no-PHI threat model and adversarial multilingual, narrative, image, document, memory, channel, and output tests |
| P0       | Enforce-capable configuration could be promoted without valid governance authorization | Environment or configuration drift could activate blocking outside the approved scope                                           | Require a dedicated profile, startup attestation, exact-artifact and exact-scope authorization, promotion record, and tested rollback |
| P1       | Shadow may be mistaken for enforcement                                                 | Users may believe a would-block prevented action                                                                                | Place visible “observe only” status in UI/reporting; preserve default shadow label                                                    |
| P1       | Shadow evidence not yet adjudicated                                                    | False positives/negatives and unmapped tools remain unknown                                                                     | Run structured evidence review before canary proposal                                                                                 |

| Priority | Risk                                                                           | Why it matters                                                                                                          | Required response                                                                                                                                           |
|----------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1       | Nurse Steward Council pending                                                  | Governance legitimacy remains founder-centered                                                                          | Name members, authority, conflicts, docket, dissent, and release linkage                                                                                    |
| P1       | Post-setup ZIPs are checksummed but not detached-signed                        | Package integrity is weaker than Harness integrity                                                                      | Sign exact manifest or document separate trust model                                                                                                        |
| P1       | Missing full SuperPowers module tree in lanes 02, 04, and 05                   | Those review-first role packs cannot truthfully install the complete referenced system                                  | Keep references non-executable; build or formally remove full-pack promise                                                                                  |
| P1       | Earlier private Nurse Leaders corpus lacks recorded installation authorization | Technical existence could be mistaken for governance consent or approved post-setup deployment                          | Keep conceptually quarantined; obtain an explicit retain, remove, or narrowly re-authorize decision                                                         |
| P1       | Human approval can launder authority the approver does not possess             | A click does not prove legal, organizational, data-use, employment, or system authority                                 | Validate source of authority; bind approval to action, target, arguments, policy, expiry, and unchanged digest; route institutional actions institutionally |
| P1       | No general post-generation delivery gate is demonstrated                       | <code>transform_tool_result</code> governs tool results, not every assistant response or delivery path                  | Map and test model-output, send, publish, channel, and delivery boundaries before claiming complete mediation                                               |
| P1       | Clinical-sounding role language may invite misuse                              | Terms such as practice, staffing, quality, or safety can prompt patient, employee, or employer data despite disclaimers | Add prominent scope labels, comprehension testing, prohibited-use tests, and qualify ambiguous launchers                                                    |

| Priority                                      | Risk                                                                            | Why it matters                                                                                                                 | Required response                                                                                                                                |
|-----------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P2<br/>current<br/>/ P0<br/>multi-user</b> | Profiles can be overread as security sandboxes                                  | Cross-user/adversarial boundaries may be weaker than assumed and become critical for mutually untrusted or institutional users | Require separate OS/container boundaries, credentials, gateways, workspaces, egress policy, and adversarial access tests where trust warrants it |
| <b>P2</b>                                     | Current public worktree is stale and locally modified                           | Future deploys could mix old canon with new content                                                                            | Reconcile through isolated worktree/branch; do not overwrite local changes blindly                                                               |
| <b>P2</b>                                     | Canary rollback is documented but not yet exercised in a dedicated live profile | Recovery behavior under real enforcement remains unproven                                                                      | Create synthetic canary profile and perform rollback drill                                                                                       |
| <b>P2</b>                                     | Argument field names are logged                                                 | Field names can sometimes reveal workflow sensitivity                                                                          | Review allowlist and minimize names where necessary                                                                                              |
| <b>P2</b>                                     | Tool-map completeness is not yet proven                                         | Unmapped tool behavior may create friction or bypass depending mode                                                            | Generate full capability inventory and path-coverage tests                                                                                       |
| <b>P2</b>                                     | Local ledger may be overstated as immutable                                     | A host administrator may rewrite both ledger and local checkpoint                                                              | Retain signed periodic roots separately; add access controls, retention, restore verification, and tamper drills                                 |
| <b>P3</b>                                     | Event ledger grows continuously                                                 | Unbounded storage and review burden can develop                                                                                | Add retention, rotation, summary, and signed terminal-root policy                                                                                |
| <b>P3</b>                                     | Institutional language can outpace evidence                                     | Architecture may be mistaken for approval or validated outcomes                                                                | Preserve Known/Assumed/Unknown/Recommended/Decide and explicit non-claims                                                                        |

# 13. Solo-founder execution sequence

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## Next 48 hours — semantic and handoff coherence

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1. Create ADR: **EDENA Risk and Autonomy Are Separate Dimensions**.
2. Replace or migrate the repository's legacy Red-autonomy model.
3. Add tests that reject any Red/high-autonomy collision.
4. Decide role-model approach: - add `educator` as a normalized identity role; or - preserve four onboarding roles and add a separate five-value `post_setup_lane`.
5. Publish an architecture erratum if the existing public architecture page implies full semantic coherence.

## Next seven days — evidence review

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1. Export a metadata-only shadow review report.
2. Categorize observed blocks by reason code and tool.
3. Sample for false positives and false negatives without exposing payloads.
4. Inventory all enabled tools/plugins/MCPs/cron/delegation/delivery paths.
5. Produce an end-to-end no-PHI threat model across intake, context, memory, attachments, tools, delegation, channels, and output.
6. Test that the post-setup user flow asks educator-versus-leader explicitly.
7. Confirm all five packages still verify after any role-schema change.

## Next 30 days — signed distribution and canary proposal

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1. Produce post-setup release manifest v1.1.
2. Bind it to a detached signature through exact-digest human authorization.
3. Create a dedicated `naio-canary` profile.
4. Restrict it to synthetic, no-PHI cases and no live connectors.
5. Run allow, unknown-tool, synthetic-sensitive, approval-denial, evaluator-failure, output-budget, and rollback tests.
6. Present the complete canary proposal to Robert; do not activate automatically.

## Before broader enforcement

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- Nurse Steward Council process is operational, not merely documented.



- Shadow and canary evidence are reviewed.
- Tool-path coverage is demonstrated.
- False-positive burden is acceptable.
- Rollback drill passes.
- Default-profile user-visible behavior is documented.
- Robert authorizes the exact signed artifact, policy, profile, scope, and rollback plan.

## 14. Decisions for Robert

| Decision                                                        | Recommendation                                                             | Why now                                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Which EDENA semantics are canon?                                | Risk colors + A0–A4 autonomy; Red = stop/escalate                          | Runtime and repository currently conflict                                              |
| How should Educator be represented?                             | Separate <code>post_setup_lane</code> ; preserve broad onboarding identity | Avoid unnecessary identity migration while enabling precise routing                    |
| Should post-setup packages be signed?                           | Yes, exact manifest detached signature                                     | Align public distribution with NAIO trust posture                                      |
| What is the status of the earlier private Nurse Leaders corpus? | Conceptually quarantined pending a separate explicit disposition           | Public role-pack authorization did not retroactively authorize the earlier integration |
| Should canary enforcement begin now?                            | No                                                                         | Shadow evidence has not been adjudicated and semantic canon is inconsistent            |
| When should Council review occur?                               | Before any broad enforcement; begin recruitment/activation now             | Founder-only governance cannot scale institutionally                                   |
| Should the default profile remain shadow?                       | Yes                                                                        | It protects normal work while evidence and semantics are reconciled                    |

## 15. Architecture acceptance criteria

Nurse AI OS may call the architecture **coherent** when:

- [ ] one canonical artifact defines risk and autonomy;
- [ ] every runtime, schema, website, quiz, package, and curriculum passes semantic validation;

- [ ] the six English post-setup lanes have an explicit user-controlled resolver and durable handoff field;
- [ ] role selection does not infer credentials or authority;
- [ ] shadow evidence is reviewed and documented;
- [ ] tool-path coverage is measured;
- [ ] no-PHI coverage is threat-modeled across pre-tool, tool, memory, media, channel, and output paths;
- [ ] post-setup distribution integrity has a deliberate trust model;
- [ ] Council status is visible;
- [ ] no-PHI and nonclinical boundaries remain unchanged.

Nurse AI OS may call the architecture **canary-enforced** only when:

- [ ] a dedicated canary profile exists;
- [ ] synthetic no-PHI tests are the only workload;
- [ ] unknown tools and evaluator failures fail closed;
- [ ] external side effects require meaningful approval;
- [ ] denial and timeout prevent execution;
- [ ] rollback has been exercised;
- [ ] Robert explicitly authorizes the exact scope.

Nurse AI OS may not call the architecture **clinically deployed** without an entirely separate institutional, legal, security, privacy, clinical-safety, human-factors, and operational-validation program.

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## 16. Confidence, limitations, and attack findings

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### Confidence

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#### High confidence

- signed release identity and signature verification;
- active plugin/version/mode;
- unit and synthetic evaluation counts;
- immutable release layout;
- live post-setup page and package integrity;
- shadow ledger chain and no-payload flag at the evidence snapshot;
- repository EDENA semantic conflict;
- SOUL role-enum/post-setup-lane mismatch.

These findings are grounded in inspected files, executable verification, Git state, or live HTTP/download checks.

Three independent read-only architecture reviews challenged implementation status, enforcement language, tool coverage, no-PHI claims, role-package completeness, isolation, signing, institutional authority, and CNO/architect readability. Their material evidence-backed corrections are incorporated in this revision. They do not constitute a security audit, clinical review, institutional validation, or Nurse Steward Council approval.

## Medium confidence

- completeness of tool-path coverage;
- practical false-positive burden;
- effectiveness of future canary enforcement under all Hermes paths;
- whether post-setup users will correctly follow two-step review.

These require structured pilot evidence rather than code inspection alone.

## Not established

- clinical safety;
- institutional suitability;
- patient or workforce outcomes;
- HIPAA compliance;
- external security certification;
- nurse-user adoption at scale;
- Council legitimacy or approval.

## Architectural attack

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The strongest argument against the current architecture is not that it lacks governance components. It has many. The strongest argument is that **a system can appear more governed than it is when doctrine, runtime, distribution, and human institutions mature at different speeds.**

That attack is valid here:

- the signed runtime semantics are ahead of the repository canon;
- the public role distribution is ahead of the durable SOUL lane handoff;
- the plugin evidence plane is ahead of enforcement authorization;
- the governance documentation is ahead of Council activation;
- the checksum discipline of role packages is behind the signing discipline of the Harness.
- the tool-call observer is ahead of end-to-end intake, memory, media, output, and delivery coverage;

- approval machinery is ahead of demonstrated approver-authority validation for institutional actions.

The correct response is not to discard the architecture. It is to make status boundaries visible and close the coherence gaps before increasing autonomy or clinical proximity.

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## 17. Final architectural position

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Nurse AI OS now has a credible technical spine:

- Hermes as the sole runtime;
- SOUL as the accountable identity layer;
- EDENA as the governance decision model;
- Florence-X as the verification and coordination discipline;
- a signed Hermes-native Harness;
- a local metadata-only evidence ledger;
- immutable releases and rollback;
- a live six-lane English post-setup distribution, including Nursing Student and Assistant, Nurse Leader, and USA-only Nurse Practitioner Complete Editions.

Its next phase is not “more agents.” It is **coherence, evidence adjudication, and human governance**.

The immediate work is to ensure that every layer tells the same truth:

- Red means stop.
- Autonomy has its own axis.
- Shadow means observe only.
- A download is not an installation.
- A role choice is not authority.
- A signature is not clinical approval.
- A nurse remains the steward.

### What would a bedside nurse at 3 AM need to know?

The system can help prepare the work. It is not authorized to take your judgment, accept patient data, or act beyond the scope you approved. Today, the new governor is watching and learning; it is not yet blocking. Before that changes, the rules, role handoff, evidence, rollback, and human authority must all agree.

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# Appendix A — Key evidence artifacts

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## Local signed Harness

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- `~/hermes/naio-harness-v2/ACTIVATION.json`
- `~/hermes/naio-harness-v2/ROLLBACK.md`
- `~/hermes/naio-harness-v2/current/evidence/release-evidence.json`
- `~/hermes/naio-harness-v2/current/evidence/release-evidence.sig`
- `~/hermes/naio-harness-v2/current/config/edena-semantics.json`
- `~/hermes/naio-harness-v2/current/src/naio_harness/runtime_gate.py`
- `~/hermes/naio-harness-v2/state/runtime-events.jsonl`

## Repository and public distribution

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- `naio-os/config/edena-policy.yaml`
- `naio-os/config/florence-x.yaml`
- `naio-os/schema/naio-soul.schema.json`
- `soul-quiz.html`
- <https://nurse-ai-os.org/governed-harness.html>
- <https://nurse-ai-os.org/post-setup/>
- <https://nurse-ai-os.org/post-setup/downloads/manifest.json>

## Publication evidence boundary

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This report is self-contained for public review. An earlier internal Harness working report informed the analysis but is **not incorporated by reference** and is not part of the public verification set. Every claim retained from that work is restated here and tied to the inspectable evidence artifacts listed above.

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# Appendix B — Live post-setup downloads

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Nursing Student and Nursing Assistant · Staff Nurse · Nurse Leader and Manager · Nurse Educator · Nurse-Connected Ally · Nurse Practitioner — USA only